

Introduction

Orthostatic hypotension (OH) is a chronic and often debilitating condition characterized by impaired autonomic regulation and an inadequate blood pressure response to postural change. Standard management relies heavily on pharmacologic therapy, including midodrine, fludrocortisone, droxidopa, and pyridostigmine, which act through vasoconstriction, plasma volume expansion, or modulation of autonomic signaling. While these agents can provide symptomatic relief, their use is frequently limited by incomplete efficacy, adverse effects, and substantial financial burden, particularly in the case of droxidopa.

Emerging evidence suggests that impaired supracardiac venous outflow may contribute to orthostatic intolerance through disruption of venous return, altered baroreflex signaling, and impaired cerebral perfusion. Endovascular interventions targeting the internal jugular and brachiocephalic veins have been associated with improvements in orthostatic hemodynamics and patient-reported symptoms.

As physiologic stability improves following intervention, an important clinical question arises: whether ongoing pharmacologic support remains necessary. In routine practice, however, there is no standardized framework to guide medication adjustment after intervention, and patients are often maintained on pre-procedural regimens despite potential improvement.

In this study, we evaluate changes in medication burden following supracardiac venous intervention and describe a **pragmatic, clinician-directed medication de-escalation framework** implemented in parallel with post-procedural monitoring. This approach was designed to reduce treatment burden while maintaining hemodynamic stability, though it was not independently studied as a therapeutic intervention.

Methods

Study Design and Population

This study represents a prospective analysis of patients with orthostatic hypotension undergoing supracardiac venous angioplasty and/or stenting. Patients were included if they had available pre- and post-intervention data on orthostatic blood pressure measurements, medication use, and clinical follow-up.

Venous Intervention

Endovascular treatment targeted the internal jugular veins and/or brachiocephalic veins and consisted of balloon angioplasty with or without stent placement. Interventions were guided by venographic and intravascular ultrasound findings, including luminal narrowing, abnormal flow

patterns, delayed contrast clearance, and/or pressure gradients consistent with impaired venous outflow.

Orthostatic Hemodynamic Assessment

Orthostatic blood pressure measurements were obtained in the supine, seated, and standing positions. Paired supine and standing systolic blood pressure (SBP) measurements obtained within a 15-minute interval were used to calculate orthostatic SBP drop (supine – standing). Pre- and post-intervention values were compared on a per-patient basis.

Medication Burden Assessment

Medication use was recorded at baseline and follow-up. To standardize comparisons across pharmacologic agents, medication burden was quantified using **Midodrine Equivalent Units (MEQ)**, incorporating doses of midodrine, fludrocortisone, droxidopa, and pyridostigmine into a unified scale. Pre-intervention medication burden was defined as the highest stable baseline regimen, and post-intervention burden was defined as the prescribed regimen at follow-up.

Medication De-escalation Framework

Following intervention, medication reduction was pursued using a **pragmatic, clinician-directed framework** intended to decrease treatment burden while maintaining hemodynamic stability.

The preferred sequence of medication discontinuation was:

1. Droxidopa
2. Pyridostigmine
3. Fludrocortisone
4. Midodrine

This sequence was selected based on pharmacoeconomic considerations, side-effect profiles, and clinical reversibility. Droxidopa was prioritized due to its high cost and variable response, while midodrine was retained longest given its rapid onset and utility as a rescue agent.

At each step, if a patient was actively taking the next medication in sequence, it was discontinued. If not, the protocol advanced to the next agent. Following medication adjustment, patients underwent:

- Orthostatic vital sign assessment (supine, seated, standing)
- Symptom evaluation (dizziness, presyncope, fatigue)

If patients experienced intolerance to medication discontinuation, defined by worsening symptoms or hemodynamic instability, the protocol was halted and a slower, individualized outpatient taper was implemented at the discretion of the treating physician.

Importantly, the present study was **not designed to formally evaluate the safety or comparative effectiveness of this de-escalation strategy**, and the framework should be interpreted as a pragmatic clinical approach rather than a validated protocol.

Outcomes

Primary outcomes included:

- Change in total MEQ from pre- to post-intervention
- Proportion of patients achieving $\geq 50\%$ reduction in MEQ

Secondary outcomes included:

- Complete medication independence (MEQ = 0 at follow-up)
- Association between medication reduction and changes in orthostatic SBP

Statistical Analysis

Continuous variables were compared using paired t-tests or Wilcoxon signed-rank tests as appropriate. Categorical outcomes were summarized as proportions. A p-value < 0.05 was considered statistically significant.

Discussion

In this study, we observed a reduction in medication burden following supracardiac venous intervention in patients with orthostatic hypotension. These findings suggest that physiologic improvements in venous outflow and orthostatic hemodynamics may reduce reliance on pharmacologic support in a subset of patients.

Current management of orthostatic hypotension often requires multi-drug regimens that are associated with substantial cost, side effects, and variable efficacy. In particular, droxidopa represents a significant financial burden, while other agents such as fludrocortisone and midodrine may contribute to supine hypertension, fluid retention, or other adverse effects. As such, reducing medication burden represents a clinically meaningful outcome beyond traditional hemodynamic measures.

To guide medication adjustment in this context, we implemented a structured, clinician-directed de-escalation framework based on cost, tolerability, and pharmacologic characteristics. This approach prioritized discontinuation of higher-cost and less predictable therapies while preserving agents with rapid onset and titratability for as long as needed.

However, the present analysis was not designed to isolate the independent effect of this de-escalation strategy. Medication reduction occurred in parallel with physiologic improvement

following intervention, and causality between the specific sequence of withdrawal and clinical outcomes cannot be established. Accordingly, the proposed framework should be interpreted as a pragmatic management strategy rather than a validated therapeutic algorithm.

These findings highlight an important shift in the management paradigm of orthostatic hypotension. From chronic pharmacologic compensation toward targeted, mechanism-based intervention with the potential for reduced medication dependence. Future prospective studies are needed to define optimal strategies for medication tapering, including the safest sequence, timing, and patient selection criteria.